OWNER 3 → OWNER 4

Integration Handoff • Red Thread Game — Epistemic Engine

1. Purpose

The Epistemic Engine controls what an NPC knows and what information the NPC is permitted to reveal during dialogue. It helps prevent dialogue generation from exposing secret or progression-locked facts.

2. Main integration method

Owner 4 should call this method for the current dialogue turn:

```
authorized_knowledge = epistemic.get_authorized_knowledge(
    state,
    suspect_id,
    presented_clue_ids=presented_clue_ids,
)
```

This returns an **AuthorizedKnowledge** object containing the NPC's knowledge boundary for that turn.

3. Important fields

Field	Meaning	Use in dialogue?
known_fact_ids	Facts assigned as known to the NPC.	No, not directly
secret_fact_ids	Facts marked as secret for the NPC.	No
available_fact_ids	Facts currently authorized by progression; may include secrets.	No
revealable_fact_ids	Facts safe to disclose in this dialogue turn.	YES
concealed_fact_ids	Authorized secret facts not released in this turn.	No
belief_ids	Identifiers of authored NPC beliefs.	Only if separately supported

4. Integration flow

1. Receive the current **GameState**.
2. Identify the NPC/suspect being interviewed.
3. Collect clue IDs presented by the player in this dialogue turn.
4. Call **get_authorized_knowledge()**.
5. Pass only **revealable_fact_ids** as factual context to dialogue generation.
6. Generate the NPC response using that filtered context.

5. Example integration

```
authorized_knowledge = epistemic.get_authorized_knowledge(
    state,
    suspect_id,
    presented_clue_ids=presented_clue_ids,
)

dialogue_fact_ids = authorized_knowledge.revealable_fact_ids

# Pass only dialogue_fact_ids to the dialogue-generation module.
```

The dialogue module can retrieve the corresponding fact content from case data using these IDs.

6. Important restrictions

• Never pass **available_fact_ids** directly to the dialogue generator.

• Never expose **secret_fact_ids** merely because the NPC knows them.

• Do not independently decide whether a fact is unlocked.

• Do not bypass the Epistemic Engine's evidence-reaction checks.

• Recalculate authorized knowledge for each dialogue turn because progression and presented evidence may change.

7. Current Owner 3 validation

Component	Status
epistemic_models.py	Implemented and reviewed
epistemic_engine.py	Implemented and reviewed
test_epistemic_engine.py	12 tests passing
Secret fact leakage regression test	Passing

8. Pending integration validation

Owner 4's dialogue module still needs to be connected to the Epistemic Engine. After integration, the team should test:

• An NPC does not reveal a secret fact before the required stage.

• An NPC reveals the correct secret when valid evidence is presented.

• An NPC does not reveal unrelated or unauthorized facts.

• Dialogue generation receives only revealable_fact_ids.

• Repeated evidence does not cause unauthorized disclosure.

Ownership: Owner 3 — Epistemic Engine **Integration responsibility:** Owner 4 — Dialogue Generation